Joseph Lavond

209-269-7175 | joelavond@gmail.com | linkedin.com/in/joelavond | github.com/JoeLavond

Education

University of North Carolina — Chapel Hill, NC
Ph.D. Statistics and Operations Research — *Aug. 2020 – May 2025*
M.S. Statistics and Data Science — *Aug. 2020 – May 2024*

California Polytechnic State University — San Luis Obispo
B.S. Statistics — *Sep. 2016 – Mar. 2020*

Experience

Data Scientist — June 2025 – Present
Red Ventures — *Fort Mill, SC*

- Authored the LLM cost roadmap for Kamino, RV's AI agent platform (15 business units, 120+ agents), and delivered its measurement and prompt-caching phases, cutting cost per model call ∼30% (∼$65K/yr)
- Built trace monitoring that prioritizes agent conversations by implicit signals (trace errors) and user feedback, classifies failures, and generates A/B-validated eval cases and recommendations; live in 3 business units
- Built ReplayOPE, an offline policy evaluation platform for contextual bandits; challengers beat production baselines in 67% of 54 analyses (3.2% median lift), with recommendations traced to production changes across 18 agents
- Fixed a 3-year missing-values bug and a 2-year policy-importance bug in offline RL training, and productionized CatBoost, ordinal encoding, and recency-based sample weighting (Replay-estimated lifts up to 10%)
- Architected a Claude Code skills marketplace with LLM security gating and SHA-pinned releases, adopted by the security org as the company-wide gate for approved skills

Graduate Data Science Intern — June 2024 – Aug. 2024
Red Ventures — *Fort Mill, SC*

- Implemented offline model-based Q-learning for website decisioning, demonstrating a typical 5% revenue-per-session lift and estimated $5M annual revenue opportunity
- Built a historical-production-data simulation pipeline for Markov Decision Process experiments, increasing simulation quantity, complexity, and robustness

Graduate Research Intern — June 2023 – Aug. 2023
Cisco — *San Jose, CA*

- Developed scalable differentially private generative-model training methods and evaluated GAN mode-collapse mitigation techniques

Graduate Info Technology Intern — June 2022 – Aug. 2022
Elevance Health — *Indianapolis, IN*

- Built a PyTorch semi-supervised Bayesian anomaly-detection model on AWS and created a PySpark SQL pipeline for 60M-row claims data

Publications

- Joseph Lavond, Minhao Cheng, and Yao Li. Feddecay: Balancing model performance and rapid personalization in federated learning with learning rate scheduling. *Journal of Computational and Graph Statistics*, 2026. Accepted
- Joseph Lavond, Minhao Cheng, and Yao Li. Trusted aggregation (TAG): Backdoor defense in federated learning. *Transactions on Machine Learning Research*, 2024

Skills

ML & Decision Systems: Contextual Bandits, Offline Policy Evaluation, Reinforcement Learning, LightGBM, CatBoost, Bayesian Inference

LLM & Agent Systems: LLM Evaluation, Trace Monitoring, Prompt Caching, Cost Attribution, MCP, AWS Bedrock

Production & Infrastructure: FastAPI, TypeScript/React, Databricks, GitHub Actions, Terraform, New Relic, CI/CD

Programming & Experimentation: Python, SQL, PySpark, Pandas, NumPy, Statistical Testing, A/B Testing